SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks50** **Duration :60 Minutes**

Annexure A

Analytical Problem Solving

Code of Conduct:

I V.Thanusri(192472265) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Thanusri

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation
(a) Explain the purpose of grey level transformation in image enhancement.
(b) Apply a negative transformation s=255-r to the pixel values:
r=[50,100,150,200]
Compute the transformed values.

2. Log Transformation
(a) Explain how log transformation enhances low-intensity pixels.
(b) Given s=clog(1+r), where c=10, compute s for:
r=[0,10,50,100]

3. Histogram Processing
(a) Explain the role of histogram equalization in contrast enhancement.
(b) Given a 3-bit image with histogram:

Intensity	**Frequency**
0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

4. Spatial Filtering (Smoothing)
(a) Explain the purpose of spatial filtering for smoothing.
(b) Apply a 3×3 averaging filter to the following region:
10 20 30 20 30 40 30 40 50
Compute the new center pixel value.

5. Spatial Filtering (Sharpening)
(a) Explain how sharpening enhances image details.
(b) Apply the Laplacian mask:
[0 -1 0}, {-1 4 -1], { 0 -1 0]
to the matrix:
10 10 10, 10 20 10, 10 10 10
Compute the output at the center.

Answers

1. Grey Level Transformation

(a) Purpose of Grey Level Transformation

Grey level transformation is an image enhancement technique used to improve the appearance of an image by changing the intensity values of pixels.

Purposes:

- Improves image contrast.
- Enhances important features.
- Converts dark or bright regions into more visible ranges.
- Helps in better image analysis and segmentation.

(b) Negative Transformation

Formula:

[
s = 255-r
]

Given:

[
r=[50,100,150,200]
]

Calculation:

For r = 50:

[
s=255-50=205
]

For r = 100:

[
s=255-100=155
]

For r = 150:

[
s=255-150=105
]

For r = 200:

[
s=255-200=55
]

Transformed Values:

Original Pixel (r)	New Pixel (s)
50	205
100	155
150	105
200	55

Result:
The negative transformation converts bright pixels into dark pixels and vice versa.

2. Log Transformation

(a) Explanation

Log transformation is used to enhance low-intensity pixels in an image.

Formula:

$$
s=c\log(1+r)
$$

It expands the values of darker pixels and compresses brighter pixel values, making hidden details visible in dark regions.

(b) Calculation

Given:

$$
c=10
$$

$$
s=10\log(1+r)
$$

Using log base 10:

For r = 0

$$
s=10\log(1+0)
$$

$$
s=10\log(1)=0
$$

For r = 10

$$
s=10\log(11)
$$

$$
s=10(1.041)=10.41
$$

For r = 50

[
s=10\log(51)
]

[
s=10(1.708)=17.08
]

For r = 100

[
s=10\log(101)
]

[
s=10(2.004)=20.04
]

Output Values

r	s
0	0
10	10.41
50	17.08
100	20.04

Result:
Low intensity values are expanded, improving visibility of dark areas.

3. Histogram Processing

(a) Role of Histogram Equalization

Histogram equalization improves image contrast by redistributing pixel intensity values.

Advantages:

- Enhances image brightness distribution.
- Makes hidden details visible.
- Improves contrast in low-quality images.

(b) Given Histogram

3-bit image:

Maximum intensity:

[
L=2^3=8
]

Intensity levels:

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels:

[
N=2+2+4+8=16
]

Step 1: Calculate Cumulative Frequency

Intensity	Frequency	CDF
0	2	2
1	2	4
2	4	8
3	8	16

Step 2: Equalization Formula

$$
s=(L-1)\times CDF/N
$$

Here:

$$
L-1=7
$$

For intensity 0:

$$
s=7(2/16)
$$

$$
s=0.875 \approx 1
$$

For intensity 1:

$$
s=7(4/16)
$$

$$
s=1.75\approx2
$$

For intensity 2:

[
s=7(8/16)
]

[
s=3.5\approx4
]

For intensity 3:

[
s=7(16/16)
]

[
s=7
]

Equalized Values

Original Intensity	CDF	Equalized Intensity
0	2	1
1	4	2
2	8	4
3	16	7

Result:
Histogram equalization increases contrast by spreading intensity values.

4. Spatial Filtering (Smoothing)

(a) Purpose of Spatial Filtering

Spatial filtering is used to reduce noise and smooth images by averaging neighboring pixel values.

Applications:

- Noise removal.
- Image preprocessing.
- Removing small variations.

(b) 3×3 Averaging Filter

Given region:

[10 20 30
 20 30 40
 30 40 50]

Averaging filter formula:

$$
New\ Pixel=\frac{Sum\ of\ all\ pixels}{9}
$$

Sum:

$$
10+20+30+20+30+40+30+40+50
$$

$$
=270
$$

Average:

$$
\frac{270}{9}=30
$$

New Center Pixel Value = 30

Result:
The averaging filter smooths the image by replacing the center pixel with the neighborhood average.

5. Spatial Filtering (Sharpening)

(a) Explanation

Sharpening enhances edges and fine details in an image.

It:

- Increases contrast around edges.
- Highlights boundaries.
- Improves image clarity.

(b) Laplacian Mask

Mask:

[0&-1&0
-1&4&-1
0&-1&0]

Image:

[10&10&10
10&20&10
10&10&10]

Center calculation:

[
Output=(0\times10)+(-1\times10)+(0\times10)
]

[
+(-1\times10)+(4\times20)+(-1\times10)
]

[
+(0\times10)+(-1\times10)+(0\times10)
]

[
= -10-10+80-10-10
]

[
=40
]

Output at Center = 40

Result:
The Laplacian filter detects edges and enhances image details.